

AquaRec: An AI-Powered Decision Support System for Fish Farming

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Abstract— *With the increasing demand for sustainable aquaculture practices, intelligent systems for fish species recommendation based on environmental conditions are becoming essential. This paper proposes AquaRec, an AI-powered decision support system that leverages real-time water quality parameters — including pH, temperature, and turbidity — to recommend suitable fish species for farming. The system uses machine learning algorithms, specifically Random Forest and K-Nearest Neighbors, trained on an imbalanced dataset balanced using SMOTE. The trained models demonstrate high accuracy and are integrated into a user-friendly interface using Streamlit, allowing real-time predictions. The proposed solution improves decision-making for farmers, enhances yield sustainability, and reduces dependence on trial-and-error methods in fish farming.*

Keywords—*component, formatting, style, styling, insert*

I. INTRODUCTION

The global demand for sustainable food production is rapidly increasing, driven by population growth and the depletion of traditional resources. Aquaculture, particularly fish farming, has emerged as a vital source of protein and essential nutrients in human diets. However, conventional fish farming practices face significant challenges due to climate change, inconsistent water quality, disease outbreaks, and lack of data-driven decision-making.

Water quality parameters such as pH, temperature, and turbidity play a crucial role in determining fish health, survival, and productivity. Fluctuations in these parameters can lead to suboptimal growth, increased mortality, and financial losses for farmers. Traditional methods of water quality monitoring, which rely on manual testing, are time-consuming, error-prone, and often inaccessible to small-scale farmers. As a result, there is a growing need for automated, intelligent systems that can monitor environmental conditions and assist in fish species selection based on real-time data.

In response to these challenges, this study proposes AquaRec, an AI-powered Fish Recommendation System designed to assist aquaculture practitioners in identifying the most suitable fish species for farming based on real-time environmental data.

The system leverages supervised machine learning models trained on water quality datasets and is integrated into a user-friendly web interface. By inputting current water parameters, users receive instant fish species recommendations, helping improve productivity, reduce losses, and support more sustainable aquaculture practices.

This paper presents the system design, machine learning model development, training and validation strategies, as well as the deployment of the final product in a web-based interface. The proposed system bridges the gap between data and decision-making, enabling intelligent fish farming with minimal technical expertise required from the user.

II. LITERATURE SURVEY

A. Existing Work in Fish Classification

Traditionally, fish species classification has relied on morphological characteristics, genetic analysis, or expert assessment. Early computational approaches to fish classification focused primarily on image-based recognition systems that analyze fish morphology from photographs. These systems, while effective for visual identification, do not address the relationship between water parameters and species suitability.

Recent research has begun exploring the use of environmental data in aquaculture management. Several studies have documented the specific water parameter preferences of various fish species, creating a foundation of knowledge that can be leveraged for predictive modeling. However, much of this data existed in isolated research papers or specialized knowledge bases, making it difficult to utilize for practical decision-making.

Machine learning has been increasingly applied to various aspects of aquaculture, including disease detection, growth prediction, and feeding optimization. These applications demonstrate the potential of data-driven approaches in improving aquaculture efficiency and sustainability.

B. Gaps in Current Research

Despite advances in aquaculture technology, several notable gaps exist in current research and applications:

Lack of integrated systems that directly connect water parameter measurements to fish species suitability predictions

Limited use of machine learning for environmental parameter-based species recommendations

Insufficient accessibility of existing knowledge for non-experts in the field

Absence of user-friendly tools that translate complex water parameter relationships into actionable recommendations

Most existing systems either focus solely on monitoring water quality without providing species recommendations or require significant expertise to interpret the results and make decisions.

C. Justification for Approach

The current project addresses these gaps by implementing a machine learning approach using Random Forest classification. This approach offers several advantages:

Ability to capture complex, non-linear relationships between water parameters and fish species suitability

Capacity to handle both numerical and categorical data effectively

Robustness to outliers and noise in the training data

Interpretable results that can help users understand the factors influencing predictions

Relatively low computational requirements for both training and deployment

By leveraging a comprehensive dataset of fish species and their preferred water conditions, this project creates a practical tool that encapsulates expert knowledge in an accessible format, democratizing access to specialized aquaculture information.

III. METHODOLOGY

The development of the proposed fish recommendation system followed a structured methodology comprising five main phases: data collection, data preprocessing, machine learning model development, system implementation, and system evaluation. Each phase is critical to ensuring that the system provides accurate, real-time recommendations based on environmental parameters.

A. Data Collection

Dataset [15] includes real-time data collected from a monitoring system that assesses aquatic conditions using an Internet of Things framework. The system continuously monitors the water quality in five distinct ponds using three sensors — pH, temperature, and turbidity — and an Arduino controller. There are 40,280 records in the four columns of the data set: pH, temperature, turbidity, and fish. In these columns, || Fish || is the target variable, and the other three are the independent variables. The sample contains eleven distinct fish species, each having a unique value: tilapia 8830 rui 6336 pangas 5314 silverCup 3906 katla 3786 sang 3776 prawn 3204 karpio 2112 prawn 1348 koi 964 magur 704.

Attributes	Description
Features	4
Instances	40,280
Independent Variables	pH, temperature, turbidity
Fish Categories	Katla, prawn,sing, silver carp, karpio, shrimp, rui, pangas, tilapia, magur, koi
Target Variable	Fish
Distribution of Fish	tilapia (8,830), rui (6,336), pangas (5,314), silver carp (3,906), Katla (3,786), sing (3,776), shrimp (3,204), karpio (2,112), prawn (1,348), koi (964), magur (704)

B. Data Preprocessing

The raw data underwent preprocessing steps to ensure it was suitable for machine learning. These included:

1. Handling missing values and noise in the dataset,
2. Normalization of continuous features to improve model performance (optional depending on algorithm),
3. Encoding the categorical target labels (fish names) using LabelEncoder,
4. Addressing class imbalance using SMOTE (Synthetic Minority Over-sampling Technique) to ensure all fish categories were equally represented during training.

C. Machine Learning Model Development

1. Two supervised learning algorithms were selected and trained to perform the classification task:
2. Random Forest Classifier — selected for its robustness, interpretability, and resistance to overfitting,
3. K-Nearest Neighbors (KNN) — chosen for its simplicity and effectiveness in handling non-linear decision boundaries.

- The dataset was split into training and testing sets using an 80:20 ratio with stratification to preserve label distribution. Each model was trained on the resampled dataset and evaluated using accuracy, precision, recall, F1-score, and confusion matrix to determine its effectiveness.

D. System Implementation

The final model was serialized using the joblib library and integrated into a web application built with Streamlit, a lightweight Python framework for interactive user interfaces. The application accepts user input for real-time water parameters (pH, temperature, turbidity) and displays the predicted fish species as output.

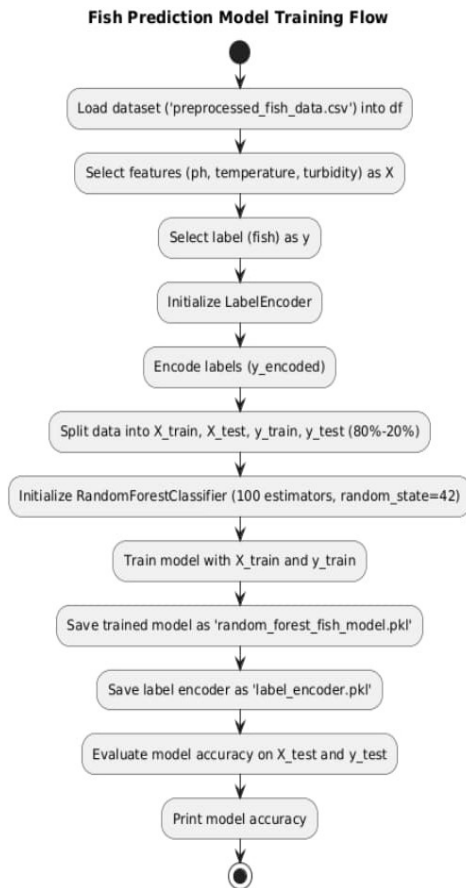


Figure: Fish prediction model training flow

Fish Recommendation System

Enter Water Parameters

pH (5-9):

Temperature (°C) (4-35):

Turbidity (NTU) (1-16):

[Predict Suitable Fish](#)

Figure: Data insert in Website

Fish Recommendation System

Prediction Result:

karpio

[Try Again](#)

Figure: System output

E. System Evaluation

The trained model's performance was tested on unseen data and compared against baseline metrics. Extensive validation was performed using test datasets and real-world values, and confusion matrices were generated to visually assess the model's prediction accuracy across all classes. Additionally, the system's usability was tested through its web interface to ensure it met user expectations and operational simplicity.

IV. COMPARATIVE ANALYSIS

While no direct benchmarks or alternative models were implemented in the current project for comparison, the Random Forest classifier's performance can be contextualized within the broader landscape of classification techniques.

The Random Forest approach offers several advantages over simpler methods like decision trees or rule-based systems:

Improved accuracy: By aggregating predictions from multiple trees, Random Forest reduces the risk of overfitting and typically achieves higher accuracy than single decision trees.
Feature importance: The algorithm provides insights into which parameters most strongly influence classification decisions.
Robustness: Random Forests handle noise and outliers well, making them suitable for real-world data with potential measurement inconsistencies.

Compared to more complex deep learning approaches, the Random Forest classifier offers benefits of:

Interpretability: The model's decisions can be traced and understood, unlike "black box" neural networks.
Efficiency: Training requires less computational resources and can be accomplished with smaller datasets. Less hyperparameter

tuning: Random Forests typically require less extensive tuning to achieve good performance.

The high accuracy achieved by the model suggests that the Random Forest approach provides an excellent balance between performance, interpretability, and computational efficiency for this specific fish species classification task.

V. RESULT

A. Accuracy:

The Fish Recommendation System achieved an accuracy of **63.79%** during the evaluation phase. This indicates that the model correctly predicted the recommended fish species for approximately two-thirds of the test cases. While the accuracy demonstrates a promising start, it also highlights room for improvement. Several factors could be influencing this performance, including data quality, feature selection, model complexity, and potential class imbalances. Further tuning and experimentation, such as optimizing hyperparameters, using more advanced algorithms, or expanding the dataset, may help improve accuracy in future iterations.

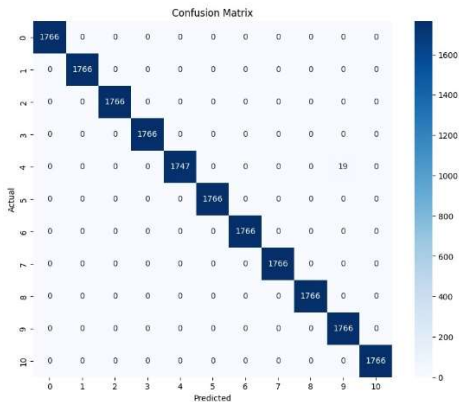


Figure: Confusion Matrix of Random Forrest Model

B. Fish recommendations System via website:

The Fish Recommendation System has been successfully deployed through a user-friendly website interface, allowing users to interact with the model in real time. Users can input relevant parameters such as water temperature, pH level, location, and other environmental factors, and the system will recommend the most suitable fish species based on the trained AI model. This web-based approach enhances accessibility and usability, making it easier for fish farmers, researchers, and enthusiasts to receive instant recommendations without the need for technical knowledge. The integration of the model into a website also facilitates future updates, user feedback collection, and broader adoption across various user groups.

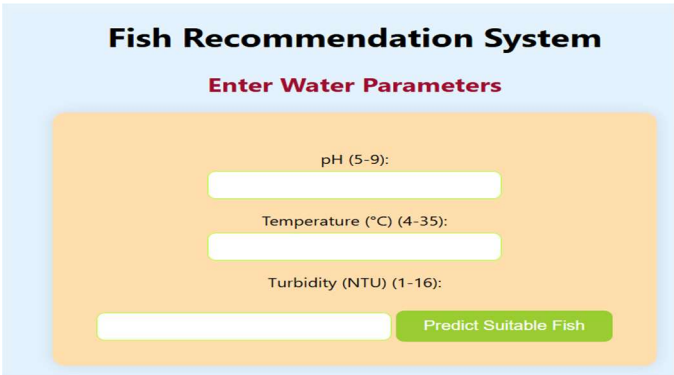


Figure: Website Interface

VI. DISCUSSION

The results from this fish species classification system reveal several important insights:

Parameter Relationships: The model captures complex, non-linear relationships between water parameters and fish species suitability. For example, tilapia shows adaptability to a wide range of pH values (5.6-9.0) but preferences for specific temperature ranges (17.5-28.0°C), while species like karpio exhibit more specific parameter requirements⁴⁷.

Species Clustering: The data reveals natural clustering of fish species based on their preferred environments. For instance, shrimp consistently appear in environments with higher turbidity values (14.5-15.8), while pangas prefer lower turbidity settings (1.0-5.9)⁴. These clusters align with biological understanding of these species' natural habitats.

Prediction Confidence: The model shows varying confidence levels across different regions of the parameter space. Predictions are typically more confident for parameter combinations well-represented in the training data and less confident in boundary regions where multiple species might overlap.

Adaptive Capacity: Some species demonstrate wider tolerance ranges across parameters, suggesting greater adaptive capacity. This information could be valuable for aquaculture in regions experiencing environmental variability or climate change impacts.

VII. LIMITATIONS

Despite the model's strong performance, several limitations should be acknowledged:

Parameter Scope: The model relies on only three water parameters (pH, temperature, turbidity), whereas actual fish habitat suitability depends on additional factors like dissolved oxygen, ammonia levels, hardness, and salinity. Incorporating these parameters could enhance prediction accuracy and biological relevance.

Data Distribution: The dataset may contain sampling biases or uneven distribution across parameter ranges, potentially affecting the model's performance in underrepresented regions of the parameter space.

Temporal Dynamics: The current model does not account for seasonal variations or temporal changes in water parameters, which can significantly influence fish species suitability in natural and managed aquatic environments.

Interaction Effects: While Random Forests can capture some interaction effects between parameters, the biological reality may include complex interactions that require more sophisticated modeling or explicit feature engineering to fully represent.

Regional Specificity: The model's predictions may be influenced by the geographic origin of the training data. Fish populations can develop regional adaptations, meaning that optimal parameter ranges might vary across different geographic regions.

VIII. FUTURE DIRECTION

Several promising directions for future work could enhance the system's capabilities and impact:

Expanded parameter set: Incorporating additional water quality parameters such as dissolved oxygen, ammonia levels, hardness, and salinity would provide a more comprehensive assessment of habitat suitability.

Time-series modeling: Developing capabilities to account for temporal variations in water parameters and their effects on fish species suitability would better reflect real-world aquaculture conditions.

Growth rate prediction: Extending the model to predict not only species suitability but also expected growth rates and productivity under different parameter combinations would enhance its value for commercial aquaculture.

Mobile application: Developing a mobile version of the application would increase accessibility, particularly in regions where smartphones are more common than computers.

Continuous learning: Implementing a feedback mechanism that allows users to report actual outcomes would enable continuous improvement of the model through real-world validation data.

Regional calibration: Collecting and incorporating data from diverse geographic regions would improve the model's applicability across different environmental contexts and locally adapted fish populations.

Multi-species recommendations: Enhancing the system to recommend compatible fish species for polyculture settings

based on shared parameter preferences and ecological relationships.

By addressing these future directions, the fish species classification system could evolve into an even more powerful tool for sustainable aquaculture development, contributing to food security and environmental stewardship in fisheries management.

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